

GABES, Tunisia

WMO: 607650

Lat: 33.877N

Lon: 10.103E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.0	5.2	-6.2	2.2	17.3	-4.2	2.6	14.2	13.0	14.9	11.7	14.5	2.6	270	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	39.8	20.9	37.0	21.1	34.9	21.6	27.1	31.2	26.4	30.6	25.7	30.0	5.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	30.2	25.1	20.3	29.5	24.2	19.2	29.0	85.7	31.2	82.3	30.6	79.0	30.1	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.7	8.5	DB	2.1	44.4	1.3	1.6	1.1	45.5	0.4	46.4	-0.4	47.3	-1.4	48.5	
			WB	0.2	28.1	1.2	1.5	-0.6	29.2	-1.3	30.1	-2.0	31.0	-2.9	32.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.5	12.0	13.0	15.8	18.8	22.2	25.7	28.5	29.1	26.8	22.9	17.4	13.2
	DBStd	6.53	2.24	2.77	2.90	2.62	2.74	2.67	2.25	1.99	2.57	2.84	2.74	2.51
	HDD10.0	13	6	4	0	0	0	0	0	0	0	0	0	3
	HDD18.3	676	197	152	91	23	1	0	0	0	0	1	50	161
	CDD10.0	3843	68	88	179	265	379	472	573	591	505	401	221	102
	CDD18.3	1461	0	2	11	37	121	222	315	332	255	143	21	1
	CDH23.3	15487	1	26	142	295	829	2128	3913	4258	2685	1077	123	9
(14) Wind	WSAvg	4.1	4.1	4.1	4.2	4.5	4.8	4.7	4.4	4.1	4.1	3.5	3.5	3.8
	PrecAvg	200	20	16	21	16	7	3	1	1	23	35	22	27
	PrecMax	428	121	92	91	122	29	16	7	18	121	266	107	201
	PrecMin	98	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	81	25	20	21	20	8	4	2	3	29	52	28	34
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.0	26.9	32.0	34.0	39.0	42.1	43.1	42.6	40.1	36.0	29.1	24.8
		MCWB	12.2	13.4	15.9	16.1	18.3	19.8	21.0	21.4	20.9	18.5	16.0	13.3
	2%	DB	19.8	22.9	27.1	29.8	33.3	37.0	39.1	38.2	36.2	32.8	26.2	21.1
		MCWB	11.6	12.5	13.3	16.0	18.0	19.7	22.2	22.1	21.7	19.2	16.1	13.0
	5%	DB	18.1	20.2	23.9	26.4	30.0	33.2	36.0	35.9	33.7	30.1	24.5	19.3
		MCWB	11.1	11.4	13.2	15.7	18.2	20.1	21.8	22.5	22.0	19.4	16.1	12.7
	10%	DB	16.8	18.2	21.2	24.1	27.2	31.0	33.8	33.8	31.4	28.1	22.9	18.1
		MCWB	10.7	10.9	13.0	15.3	17.7	20.3	22.2	22.9	22.9	20.3	15.8	12.5
	0.4%	WB	14.7	15.2	17.7	20.6	22.8	25.7	27.9	28.3	27.7	25.1	21.1	16.9
		MCDB	17.2	19.7	23.4	24.9	28.0	30.1	31.5	32.7	31.4	27.8	24.1	18.9
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	13.7	14.2	16.5	19.1	21.7	24.6	26.7	27.4	26.5	23.9	19.5	15.5
		MCDB	15.9	18.5	20.5	23.3	26.3	29.4	30.9	31.6	30.1	27.3	22.6	18.1
	5%	WB	12.9	13.2	15.6	18.1	20.9	23.7	25.8	26.6	25.6	22.9	18.2	14.4
		MCDB	15.8	17.3	19.7	22.1	25.3	28.4	30.3	30.9	29.5	26.4	21.9	17.6
	10%	WB	12.0	12.4	14.9	17.2	20.1	22.8	25.1	26.0	24.8	21.9	17.0	13.3
		MCDB	15.5	16.6	18.8	21.3	24.4	27.6	29.9	30.3	28.9	25.6	20.8	17.0
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	10.0	10.0	9.5	9.1	9.3	9.7	9.4	9.0	9.6	9.9	9.5
		MCDBR	11.6	13.1	15.3	14.8	15.1	14.8	15.3	14.6	13.5	13.3	12.3	11.0
		MCWBR	6.5	6.4	6.6	6.3	6.0	5.9	6.6	6.6	6.5	5.9	6.4	6.4
	5% WB	MCDBR	9.2	9.8	9.7	9.6	8.8	9.0	9.0	8.6	8.7	8.5	9.0	9.4
		MCWBR	6.4	6.0	5.5	5.7	5.0	5.6	6.1	6.0	5.9	5.5	6.5	6.5
(40) Clear-Sky Solar Irradiance	taub		0.369	0.398	0.484	0.536	0.553	0.510	0.537	0.525	0.530	0.482	0.425	0.380
	taud		2.241	2.144	1.912	1.813	1.801	1.928	1.882	1.917	1.910	1.992	2.093	2.202
	Ebn at Noon		821	836	788	767	758	789	765	766	738	736	745	779
	Edh at Noon		110	134	184	212	217	191	199	189	182	155	127	108
(44) All-Sky Solar Radiation	RadAvg		3.09	3.99	5.11	6.20	6.99	7.61	7.63	6.81	5.48	4.27	3.28	2.78
	RadStd		0.14	0.17	0.25	0.28	0.35	0.24	0.20	0.19	0.21	0.20	0.13	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-1.35	-2.02	+2.74	-0.84	-0.99	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+0.22	N/A	N/A	N/A	-0.38	N/A	N/A	N/A	+103	+60

Nomenclature: See separate page